

BCAC 0027: SOFTWARE ENGINEERING AND TESTING

Objective: The aim of the subject is to assist the student in understanding the basic theory of software engineering, and to apply these basic theoretical principles to a group software development project.

Credits: 04

L-T-P: 4-0-0

Module No.	Content	Teaching Hours
I	Introduction: Introduction to Software Engineering, Software characteristics, Software Crisis, Software Engineering Process. Software Development Life Cycle (SDLC) Models: Waterfall, Incremental, Iterative Enhancement, Prototype, RAD and Spiral Models. Software Requirements Engineering: Types of Requirements, Requirement Elicitation Techniques Like Interviews, FAST & QFD, Use case Approach, Requirements Analysis Using DFD, Data Dictionaries & ER Diagrams, Requirements Documentation, and SRS. Software Project Planning: Size Estimation like Lines of Code & Function Count, Cost. Estimation Models: COCOMO (Basic, Intermediate) Software Design: Cohesion & Coupling, Classification of Cohesion & Coupling, Function Oriented Design, Object Oriented Design, Structure chart. Coding: Characteristics of Coding and Coding style.	26
II	Software Metrics: Software Measurements, Token Count, Halstead Software Measures. Software Reliability & Quality: Introduction of Mc Call's & Boehm's Quality Model, Capability Maturity Models Software Reliability Models: Basic Execution Time Model. Software Testing: Testing Fundamentals: Test Case Design, Black Box Testing Strategies, White Box Testing, Unit Testing, Integration Testing, System Testing. Introduction to Automation Testing and Testing Tools: Automated Testing Process, Framework for Automation Testing, Introduction to Automation Testing Tool. Software Maintenance: Maintenance Process Maintenance models: Belady and Lehman Model, Boehm Model Regression Testing, Software Configuration Management; Implementation, Introduction to Reengineering and Reverse Engineering. Software Risk Management: Risk Identification and Risk Analysis	26

Text Book:

- P Jalote, (2006), "*Integrated Approach to Software Engineering*", Narosa Book Distributors Pvt. Ltd, New Delhi, 3rd edition.

Reference Books:

- K. K. Aggarwal & Yogesh Singh, (2008), "*Software Engineering*", New Age International, 3rd edition.
- R. S. Pressman, (2001), "*Software Engineering – A Practitioner's Approach*", McGraw Hill Int., 5th edition.
- Stephen R. Schach, (1996), "*Classical & Object Oriented Software Engineering*", IRWIN.
- James Peter, W. Pedrycz, "*Software Engineering: An Engineering Approach*", John Wiley & Sons.
- I. Sommerville, (2004), "*Software Engineering*", Addison Wesley, New Delhi, 7th edition.

Focus: This course focuses on Employability under CO2, CO3, CO4, CO5 and CO7.

Outcome: After the completion of the course, the student will be able to:

CO1: Understand the basic concepts of software engineering.

CO2: Apply software processes to solve real world problems.

CO3: Estimate the cost, effort and schedule of software using COCOMO Model.

CO4: Analyze the software design techniques (structure chart, SDM, sequence diagram).

CO5: Develop the test cases to validate the software.

CO6: Understand the basic models of software Quality and maintenance.

CO7: Automate the software testing using Selenium and TestNG.

Mapping of Course Outcomes (Cos) with Program Outcomes (Pos) and Program Specific Outcomes (PSOs):

Cos	Pos/PSOs
CO1	P01,P07/PS01
CO2	P02,P03/PS04
CO3	P02,P011/PS03
CO4	P03,P010/PS04
CO5	P03,P07/PS01
CO6	P05,P012/PS02
CO7	P03,P010/PS04